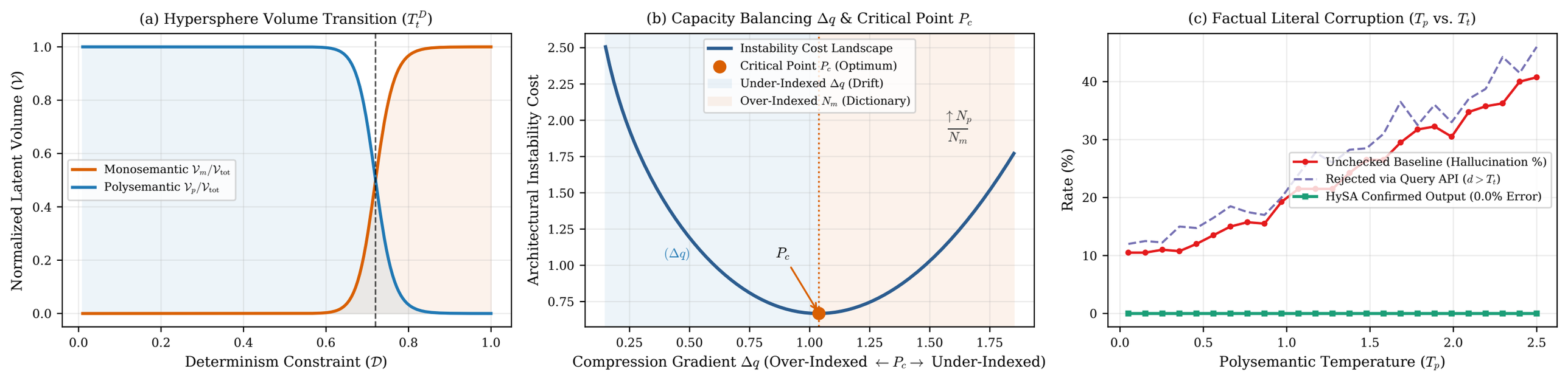
(a) Hypersphere Volume Transition (T_t^D)
Normalized Latent Volume ($\mathcal{V}$)
Monosemantic $\mathcal{V}_m/\mathcal{V}_{\text{tot}}$
Polysemantic $\mathcal{V}_p/\mathcal{V}_{\text{tot}}$
Determinism Constraint ($\mathcal{D}$)
(b) Capacity Balancing Δq & Critical Point P_c
Architectural Instability Cost
Instability Cost Landscape
Critical Point P_c (Optimum)
Under-Indexed Δq (Drift)
Over-Indexed N_m (Dictionary)
$\uparrow \frac{N_p}{N_m}$
(Δq)
P_c
Compression Gradient Δq (Over-Indexed $\leftarrow P_c \rightarrow$ Under-Indexed)
(c) Factual Literal Corruption (T_p vs. T_t)
Rate (%)
Unchecked Baseline (Hallucination %)
Rejected via Query API ($d > T_t$)
HySA Confirmed Output (0.0% Error)
Polysemantic Temperature (T_p)